5.6.2

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September 27, 2025

Question

Find $\mathbf{A}^2 - 5\mathbf{A} + 6\mathbf{I}$ if

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} \tag{1}$$

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,
We known that

$$\|\mathbf{A} - \lambda \mathbf{I}\| = 0 \tag{2}$$

$$\left\| \begin{pmatrix} 2-\lambda & 0 & 1 \\ 2 & 1-\lambda & 3 \\ 1 & -1 & -\lambda \end{pmatrix} \right\| = 0 \tag{3}$$

$$\lambda^3 - 3\lambda^2 + 4\lambda - 3 = 0 \tag{4}$$

using Cayley-Hamilton theorem

$$\mathbf{A}^3 - 3\mathbf{A}^2 + 4\mathbf{A} - 3\mathbf{I} = 0 \tag{5}$$

$$\mathbf{A}^2 - 3\mathbf{A} + 4\mathbf{A} = 3\mathbf{A}^{-1} \tag{6}$$

$$\tag{7}$$

Theoretical Solution

$$\mathbf{A}^2 - 5\mathbf{A} + 6\mathbf{I} = 3\mathbf{A}^{-1} - 2\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{pmatrix} \tag{8}$$

C Code

```c
   #include<stdio.h>
int main(){
    int arr[3][3];
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    int arr2[3][3]={0};
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            for(int z=0;z<3;z++){
                arr2[i][j]+=arr[i][z]*arr[z][j];
            }
        }
    }
```

C Code

```
for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            //printf("%d",arr2[i][j]);
            if(i==j){
                printf("%d ",arr2[i][j]-5*arr[i][j]+6);
            }
            else{
                printf("%d ",arr2[i][j]-5*arr[i][j]);
            }
        }
        printf("\n");
    }
    return 0;
}
```